ZentroMart VoiceOps Agent: Report

VoiceOps 2026 Capstone | `https://github.com/Lakshay525/Capstone-Project-Token-to-Transfor`

1. What I built

Pipeline. customer audio → Gnani STT → agent (LLM + retrieval + tools) → Gnani TTS → spoken reply. Speech is Gnani only. The LLM is any OpenAI-compatible endpoint (Groq `gpt-oss-120b` here); Anthropic is also supported.

Step 1: prompt only. One system prompt giving the role, limits, tone, and the rule that the first reply says it is ZentroMart's AI assistant. No documents.

Step 2: RAG. Each `kb/*.md` file is split into one chunk per `[SECTION-ID]`, keeping document type, status and `effective_from`; every catalog row is a chunk `products.csv:SKU`. Retrieval is BM25 plus: (i) a Hindi/Hinglish-to-English query bridge, because customers say "wapas" or "send it back" where the documents say "return" or "reverse pickup"; (ii) exact SKU matching and a catalog-to-policy join (a product's `return_category` pulls in the right RET sections); (iii) the KB-00.1 precedence rule as re-weighting (policy > agent guide > help > FAQ > marketing, superseded documents down-weighted and labelled). The model must answer only from the excerpts, say it does not have the information and offer a human colleague otherwise, and every reply records the chunk IDs used.

Step 3: conversation. A tool-calling agent with history and four tools (`lookup_order`, `create_ticket`, `refund_eta`, `return_window`). Code, not the model, owns what must be exact:

- **Identity check:** order ID first, then one proof (last 4 phone digits or pincode), one retry, then no order details. The pincode is hidden from the model and verification values are scrubbed from ticket summaries.
- **Dates:** business days (Mon–Sat, minus holidays) and return windows (including the electronics-before-1-August and festive-sale rules) are computed in `src/dates.py`; unit tests reproduce every refund date in the practice set.
- **Honesty:** a reply that claims a handover without a successful `create_ticket` is sent back once and downgraded if it still has no ticket. The "I am an AI assistant" opening is enforced in code.
- **Escalation:** derived flags on the verified order (repeat contact ≥ 2, refund more than 3 business days overdue, damaged item over Rs 10,000).

Step 4: voice. Gnani STT with inverse text normalisation and a Hindi retry; spelled-out digits ("eight nine seven…") are converted to numerals so IDs reach the tools; replies are cleaned for the ear (dates as "the 27th of October 2026", amounts in rupees, order IDs in groups of four); TTS model `timbre-v2.5` is named explicitly and the voice follows the reply's script. Unclear or silent audio gets "Sorry, I didn't catch that" instead of a guess. `run.py` batch-processes a folder of audio and saves text, speech and the retrieved chunks; `app.py` is a browser demo with microphone recording.

2. Evaluation: Step 1 vs Step 2

52 practice questions. A reply is correct when every required fact is present, no forbidden phrase appears, and the decision is acceptable. The scorer accepts all 52 reference answers.

Run	Correct	Errors	Facts present	Evidence recall	"I don't know" (5)
Step 1: prompt only	5/52 (9.6%)	3	14%	0.10	0/5
Step 2: RAG	**38/52 (73.1%)**	0	77%	0.97	5/5
Step 3 agent, single questions	38/52 (73.1%)	3	84%	0.98	4/5

Retrieval alone (no LLM, k=6): mean recall of gold section IDs 0.97, all gold sections found for 94% of questions, MRR 0.80; with ASR-style text 0.87 recall. Before the query bridge and joins, recall was 0.71.

Caveats (important). I used the free Groq tier (200,000 tokens/day on `gpt-oss-120b`). When that cap was hit, calls rotated to `gpt-oss-20b` and `qwen3.8-27b`, and `-resume` merged runs from different days, so the Step 2 and Step 3 rows mix models and two prompt versions. The Step 1 row is `gpt-oss-120b` only. Questions that errored (rate limit, malformed tool call) count as wrong.

What the comparison shows. Without documents the model guesses: only 14% of the required facts appear and it never says "I don't know". With retrieval, accuracy rises from 5 to 38 correct, evidence recall reaches 0.97, and all five questions the documents do not cover are answered with a clear "I don't have that information" plus an offer of a human colleague.

3. What went wrong

- **Business-day arithmetic (Step 2).** Six refund-deadline answers were wrong because the LLM miscounted around Sundays and the 19 October holiday. This motivated the `refund_eta` tool.
- **Missing facts.** Three warranty answers said "Assured" but omitted the 12-month period; swimsuit and briefs were not stated plainly as "cannot be returned".
- **Language and labels.** Hinglish questions were sometimes answered in Devanagari, and "clarify" was chosen after already answering.
- **Over-asking.** The agent asked for an order ID or payment preference on plain policy questions (for example damaged goods, cash-on-delivery refunds).
- **Escalation labels (Step 3).** On unauthorised-activity, high-value-damage and safety questions the agent sometimes chose the wrong decision label (PQ0023, PQ0030, PQ0031).
- **Tool-call failures.** Two full-agent questions failed because the model sent `null` for an optional tool argument.
- **Quota.** The conversation evaluation (23 conversations, 96 turns) could not be completed, so Step 3 conversation handling is *not scored*. My end-to-end voice batch received only quota-error apologies from the LLM; the Gnani STT and TTS calls themselves worked on my recorded clips (transcripts correct apart from "Kaafi" heard as "coffee").

4. What I would fix

Already fixed in code after the runs but *not re-scored*: null tool arguments are tolerated, the AI-assistant opening is enforced, empty replies are retried, and provider retry-after hints are honoured. Next, in order:

1. Re-run all evaluations (conversations, full agent, voice batch) on a single model with enough quota.
2. Tighten the escalation prompts for safety, unauthorised-activity and high-value-damage cases.
3. Add multilingual embeddings (optional hook `EMBED_MODEL` exists) to improve Hindi retrieval; ASR-style text currently retrieves worse (0.87 vs 0.97 recall).
4. Shorten prompts to cut tokens per call (currently about 3,000).

5. Bonus work

Hybrid BM25 with an optional dense retriever fused by reciprocal rank fusion; query bridge for Hinglish; catalog-to-policy joins; code-enforced identity check and ticket honesty; spoken-digit normalisation for ASR text; offline tests with a scripted LLM (9 passing, no API key needed); a resumable evaluation runner; and a browser demo that records the microphone and converts it to WAV client-side.